

1.1 Some History

The Registry was introduced as a way out of the “INI files mess” that plagued early versions of Windows. Configuration information regarding the operating system, application files, hardware, and so on were stored in INI files scattered all over the system. For example, when an application is installed, it needs to know various parameters regarding the operating system and the available hardware. Much of this information is common to other applications as well. However, if there was any change in the hardware or operating system software, updating the application’s INI files used to be hell.

The Registry was conceived as a way out of this madness. It is a hierarchical database that stores information regarding the hardware, operating system software, application software, users preferences and system preferences. The hierarchy separates the machine information, user information and the software information into logically discrete units that can be manipulated with greater ease. Any changes made by users, the operating system, newly installed software, and hardware are all updated and reflected in the Registry.

During system startup, Windows uses the registry to get configuration data regarding the hardware, software and other peripherals that have been configured. It also maintains a backup copy of the registry with which the system had successfully started up last. Hence, if it encounters a problem with the current startup, it can then use the configuration information from the backup to attempt to start the system. Some of the information stored in the registry is specific to the particular startup session only, and is mostly user-specific. When a user installs new software, Windows will either make the installation user-specific or system-wide. In some software, the user is given the option to select whether the installation should or not be available to everyone else who uses the system. In other cases, the software will install itself system-wide and save user specific information separately, making entries in the registry that are both user-specific as well as system-wide.